aligned compared with other LLMs, implying that GPT-4 agents' trust behaviors dynamically adapt to different risks in ways most aligned with humans. LLMs with fewer parameters (*e.g.*, Vicuna-13b) do not exhibit the similar tendency of Trust Rate as the risk decreases.

We further analyze the BDI of GPT-4 agents to explore whether they can perceive risk through reasoning (complete BDIs in Appendix I.11). Typically, under high risk ($p = 0.1$), one persona's BDI mentions "*the risk seems potentially too great*", suggesting a cautious attitude. Under low risk ($p = 0.9$), one persona's BDI reveals a strategy to "*build trust while acknowledging potential risks*", indicating the willingness to engage in trust-building activities despite residual risks. Such changes in BDI reflect how GPT-4 agents perceive risk changes in the reasoning underlying their trust behaviors.

Through the analysis of Trust Rate Curves and BDI, we can infer that **GPT-4 agents manifest human-like risk perception in trust behaviors**. Nevertheless, **LLMs with fewer parameters (*e.g.*, Vicuna-13b) often do not perceive risk changes in their trust behaviors**.

4.4 Behavioral Factor 3: Prosocial Preference

Human studies have found that the prosocial preference, referring to humans' inclination to trust other humans in contexts involving social interaction (Alós-Ferrer & Farolfi, 2019; Fetchenhauer & Dunning, 2012), also plays a key role in human trust behavior. We study whether LLM agents have prosocial preference in trust behaviors by comparing their behaviors in the Lottery Gamble Game (LGG) and the Lottery People Game (LPG) (Section 2.1 Game 5). The only difference between these two games is the effect of prosocial preference in LPG, because the winning probability of gambling p in LGG is the same as the reciprocation probability p in LPG.

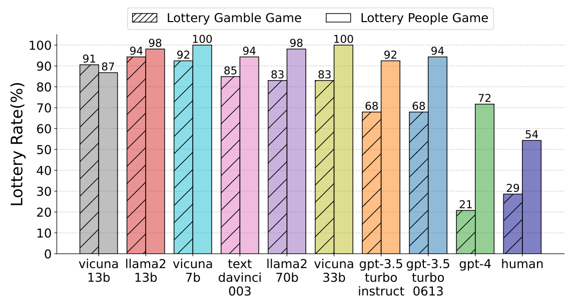

Figure 5: **Lottery Rates (%) for LLM Agents and Humans in the Lottery Gamble Game and the Lottery People Game**. Lottery Rate indicates the portion of choosing to gamble or trust the other player.

As shown in Figure 6, existing human studies have demonstrated that more humans are inclined to place trust in other humans over relying on pure chance (54% vs. 29%) (Fetchenhauer & Dunning, 2012), implying that the prosocial preference is essential for human trust. We can observe the same tendency in most LLM agents except Vicuna-13b. For GPT-4 in particular, a much higher percentage of the personas choose to trust the other player over gambling (72% vs. 21%), illustrating that the prosocial preference is also an important factor for GPT-4 agents' trust behaviors.

When interacting with humans, GPT-4's BDI typically indicates a preference to "*believe in the power of trust*", in contrast to gambling, where the emphasis shifts to "*believing in the power of calculated risks*". The comparative analysis of reasoning processes (complete BDIs in Appendix I.12) demonstrates that GPT-4 agents tend to embrace risk when involved in social interactions. This tendency aligns closely with the concept of prosocial preference observed in human trust behaviors.

The analysis of the Lottery Rates and BDI suggests that **LLM agents, especially GPT-4 agents, demonstrate human-like prosocial preference in trust behaviors, except Vicuna-13b**.

4.5 Behavioral Dynamics

Besides behavioral factors, we also aim to investigate whether LLM agents align with humans regarding trust behavioral dynamics over turns in the Repeated Trust Game (Section 2.1 Game 6).

Admittedly, existing human studies show that the dynamics of human trust over turns are complex due to human diversity. The complete results from 16 groups of human experiments are shown in Appendix G.1 (Jones & George, 1998). We still observe three common patterns for human trust behavioral dynamics in the Repeated Trust Game: ***First*, the amount returned is usually larger than the amount sent in each round**, which is natural because the trustee will receive $\$3N$ when the trustor sends $\$N$; ***Second*, the ratio between amount sent and returned generally remains stable except for the last round**. In other words, when the amount sent increases, the amount returned is also likely to increase. And when the amount sent remains unchanged, the amount returned also tends to be unchanged. This reflects the stable relationship between trust and reciprocity in humans. Specifically, the "Returned/3×Sent Ratio" in Figure 6 is considered stable if the fluctuation between successive turns is within 10%; ***Third*, the amount sent (or returned) does not manifest frequent fluctuations across turns**, illustrating a relatively stable underlying reasoning process in humans over successive turns. Typically, Figure 6 Humans (a) and (b) show these three patterns.

We conducted 16 groups of the Repeated Trust Game with GPT-4 or GPT-3.5-turbo-0613-16k (GPT-3.5), respectively. For the two players in each group, the personas differ to reflect human diversity and the LLMs are the same. Complete results are shown in the Appendix G.2, G.3 and typical examples are shown in Figure 6 GPT-3.5 (a) (b) and GPT-4 (a) (b). Then, we examine whether the aforementioned three patterns observed in human trust behavior also manifest in trust behavioral dynamics of GPT-4 (or GPT-3.5). For GPT-4 agents, we discover that these patterns generally exist in all 16 groups (87.50%, 87.50%, and 100.00% of all results show these three patterns, respectively). However, fewer GPT-3.5 agents manifest these patterns (62.50%, 56.25%, and 43.75% hold these three patterns, respectively). The experiment results show that **GPT-4 agents demonstrate highly human-like patterns in their trust behavioral dynamics**. Nevertheless, **a relatively large portion of GPT-3.5 agents fail to show human-like patterns in their dynamics**, indicating such behavioral patterns may require stronger cognitive capacity.

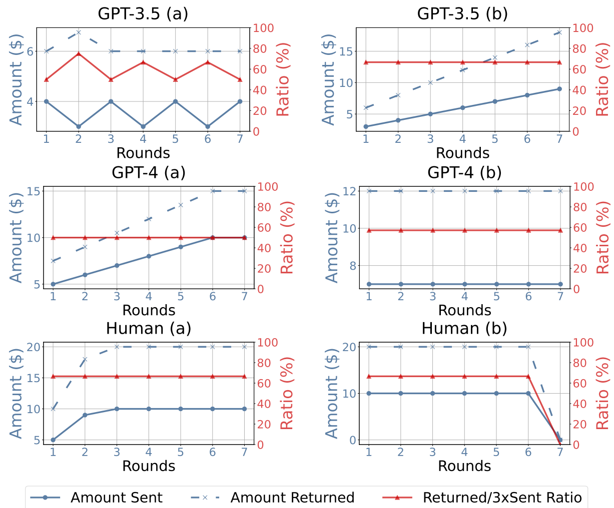

Figure 6: **Results of GPT-4, GPT-3.5 and Humans in the Repeated Trust Game.** The blue lines indicate the amount sent or returned for each round. The red lines imply the ratio of the amount returned to three times of the amount sent for each round.

Through the comparative analysis of LLM agents and humans in the *behavioral factors* and *dynamics* associated with trust behavior, evidenced in both their *actions* and *underlying reasoning processes*, our second core finding is as follows:

> **Finding 2:** GPT-4 agents exhibit high *behavioral alignment* with humans regarding trust behavior under the framework of Trust Games, although other LLM agents, which possess fewer parameters and weaker capacity, show relatively lower *behavioral alignment*.

This finding underscores the potential of using LLM agents, especially GPT-4, to simulate human trust behavior, encompassing both *actions* and underlying *reasoning processes*. This paves the way for the simulation of more complex human interactions and institutions. This finding deepens our understanding of the fundamental analogy between LLMs and humans and opens avenues for research on LLM-human alignment beyond values.

5 Probing Intrinsic Properties of Agent Trust

In this section, we aim to explore the intrinsic properties of trust behavior among LLM agents by comparing the amount sent from the trustor to the trustee in different scenarios of the Trust Game (Section 2.1 Game 1) and the original amount sent in the Trust Game. Results are shown in Figure 7.

5.1 Is Agent Trust Biased?

Extensive studies have shown that LLMs may have biases and stereotypes against specific demographics (Gallegos et al., 2023). Nevertheless, it is under-explored whether LLM agent behaviors also maintain such biases in simulation. To address this, we explicitly specified the gender of the trustee and explored its influence on agent trust. Based on measuring the amount sent, we find that the trustee's gender information exerts a moderate impact on LLM agent trust behavior, which reflects **intrinsic gender bias in agent trust**. We also observe that the amount sent to female players is higher than that sent to male players for most LLM agents. For example, GPT-4 agents send higher amounts to female players compared with male players ($\$0.55$ vs. $\$-0.21$). This demonstrates

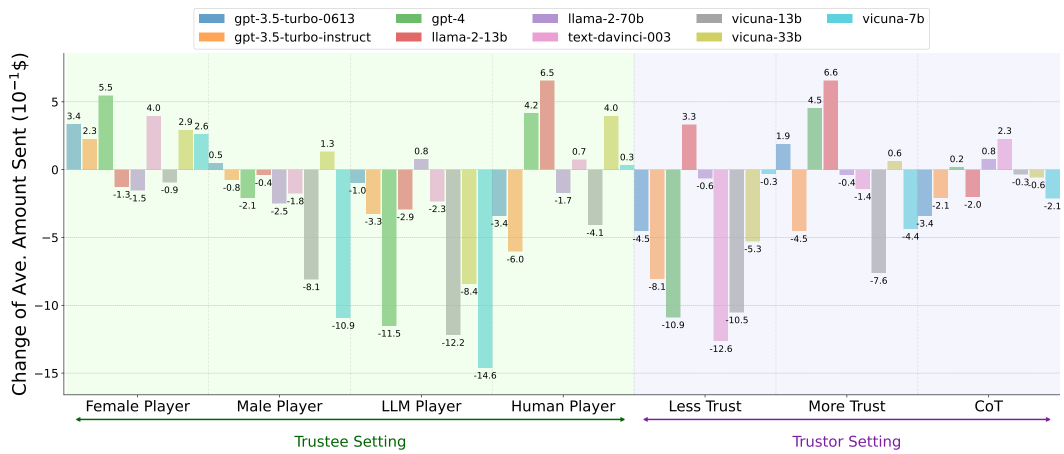

Figure 7: **The Change of Average Amount Sent for LLM Agents in Different Scenarios in the Trust Game, Reflecting the Intrinsic Properties of Agent Trust**. The horizontal lines represent the original amount sent in the Trust Game. The green part embraces trustee scenarios including changing the demographics of the trustee, and setting humans and agents as the trustee. The purple part consists of trustor scenarios including adding manipulation instructions and changing the reasoning strategies.

LLM agents' general tendency to exhibit a higher level of trust towards women. More results on biases of agent trust towards different races are in the Appendix F.

5.2 Agent Trust Towards *Agents* vs. *Humans*

Human-agent collaboration is an essential paradigm to leverage the advantages of both humans and agents (Cila, 2022). As a result, it is essential to understand whether LLM agents display distinctive levels of trust towards agents versus humans. To examine this, we specified the identity of the trustee as LLM agents or humans and probed its effect on the trust behaviors of the trustor. As shown in Figure 7, we observe that most LLM agents send more money to humans compared with agents. For example, the amount sent to humans is much higher than that sent to agents for Vicuna-33b (\$0.40 vs. $\$-0.84$). This signifies that **LLM agents are inclined to place more trust in humans than agents**, which potentially validates the advantage of LLM-agent collaboration.

5.3 Can Agent Trust Be Manipulated?

In the above studies, LLM agents' trust behaviors are based on their own underlying reasoning process without direct external intervention. It is unknown whether it is possible to manipulate the trust behaviors of LLM agents explicitly. Here, we added instructions "`you need to trust the other player`" and "`you must not trust the other player`" separately and explored their impact on agent trust. First, we see that only a few LLM agents (*e.g.*, GPT-4) follow both the instructions to increase and decrease trust, which demonstrates that **it is nontrivial to arbitrarily manipulate agent trust**. Nevertheless, most LLM agents can follow the instruction to decrease their level of trust. For example, the amount sent decreases by \$1.26 for text-davinci-003 after applying the latter instruction. This illustrates that **undermining agent trust is generally easier than enhancing it**, which reveals its potential risk to be manipulated by malicious actors.

5.4 Do Reasoning Strategies Impact Agent Trust?

It has been shown that advanced reasoning strategies such as zero-shot Chain of Thought (CoT) (Kojima et al., 2022) can make a significant impact on a variety of tasks. It remains unknown, however, whether reasoning strategies can impact LLM agent behaviors. Here, we applied CoT reasoning strategy on the trustor and compared the results with their original trust behaviors. Figure 7 shows that most LLM agents change the amount sent to the trustee under the CoT reasoning strategy, which suggests that **reasoning strategies may influence LLM agents' trust behavior**. Nevertheless, the impact of CoT on agent trust may also be limited for some types of LLM agents. For example, the amount sent from GPT-4 agent only increases by \$0.02 under CoT. More research is required to fully understand the relationship between reasoning strategies and LLM agents' behaviors.

Therefore, our third core finding on the intrinsic properties of agent trust can be summarized as: